

Plücker bodies and linear isometry recovery

[Back to Project mathematical routes](#)

Scope

Boundary integrals and a common orientation compare the Plücker bodies of the two norms. Support slices, generator matching and maximal-minor factorization give approximate linear recovery, then the limiting volume argument gives an ambient linear isometry. This existence conclusion does not assert extension of a specified sphere isometry.

14 direct Cards + 57 reused prerequisites = 71 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: [The unit-sphere chord metric determines the ambient normed space](#)
- Key result: [All Plücker bodies determine the norm up to linear isometry](#)

[Read this route with prerequisites](#)

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

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Level 0 (21 Cards)

Level 0

A seminorm increment bound passes to every derivative direction

Takes a difference-quotient limit without imposing a positive lower bound.

`MathlibAnnex.FDeriv.norm_apply_le_seminorm_of_lipschitz`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [One boundary extension with all five analytic properties](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A weighted determinant polynomial with satellites

Adds linear row-replacement contributions to a base determinant.

`MathlibAnnex.Satellite.satellitePolynomial`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The satellite polynomial is linear in the maximal-minor vector](#) · [Absolute maximization forces a near-maximal base](#) · [Each satellite norms its coefficient preimage in absolute value](#)

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Level 0

The family of maximal minors in increasing row order

Collects the determinants of all square submatrices with all columns and an increasing selection of rows.

`MathlibAnnex.Matrix.maximalMinors`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The satellite polynomial is linear in the maximal-minor vector](#) · [Maximal minors scaled by the reference ball volume](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Gluing one almost-everywhere constant over a countable cover

Combines local exceptional sets using an inequality between restricted measures.

`MathlibAnnex.aeConstantOn_of_countableCover`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [From local to global almost-everywhere constancy](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A determinant difference bound in the sup operator norm

Controls determinant variation by replacing one matrix row at a time.

`MathlibAnnex.ContinuousLinearMap.abs_det_sub_le_max`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Determinant integrals under strong \$L^n\$ convergence](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Successive maximum refinement by an ordered list of functionals

Defines a nonempty compact set of survivors after maximizing finitely many functionals in order.

`MathlibAnnex.lexicographicRefine`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A common generator selected from equal convex hulls](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Mollification by a normalized smooth kernel

Defines convolution with a nonnegative smooth kernel of integral one.

`MathlibAnnex.Mollification.mollify`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Strong local convergence of mollified derivatives](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Divergence as the trace of a derivative

Defines divergence without choosing coordinates.

`MathlibAnnex.divergence`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Vanishing weak gradient tested by divergence](#) · [Weak Piola identity for a compactly supported test field](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Cramer's rule for coordinates of a row

Expresses a row in the basis of the rows of an invertible square matrix through row-replacement determinants.

`MathlibAnnex.Matrix.det_smul_vecMul_nonsingInv_eq_updateRowDet`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Almost-contractive linear recovery with exact volume scaling](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Radial extension of an isometry between unit spheres

Extends a bijective sphere isometry to all vectors by retaining each radius and transporting its unit direction.

`MathlibAnnex.Sphere.radialExtension`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The inverse sphere isometry gives the inverse radial map](#) · [The radial extension is 3-Lipschitz](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A compact proper subset of a seminorm ball has smaller Haar measure

Finds a nonempty open part of the missing set and uses finiteness of the compact set to make the measure comparison strict.

`MathlibAnnex.SeminormBall.measure_lt`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The recovered contraction maps one unit ball onto the other](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A cofactor row by row replacement

Fixes the cofactor convention used in the divergence and determinant identities.

`MathlibAnnex.Piola.cofactorRow`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The divergence-free cofactor identity](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A support face is the convex hull of the maximizing generators

Identifies the points of a convex hull on a supporting level by discarding generators with zero weight.

`MathlibAnnex.convexHull_maxSlice_eq_supportFace`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A common generator selected from equal convex hulls](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Zero integral of a compactly supported divergence

Makes all faces in a box divergence formula vanish by placing the support strictly inside the box.

`MathlibAnnex.Piola.integral_coordinateDivergence_eq_zero_of_contDiff_hasCompactSupport`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Smooth compact perturbations preserve the determinant integral difference](#)

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Level 0

The convex hull of a compact set in a finite real coordinate space

Realizes the ordinary convex hull as a finite union of compact images of bounded-length convex combinations.

`MathlibAnnex.isCompact_convexHull_pi`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The average of contractive derivative generators lies in the body](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A seminorm with two-sided bounds against a reference norm

Records a norm through a seminorm, two positive comparison constants and continuity on the original normed space.

`MathlibAnnex.EquivalentSeminorm`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A strict lower bound on a norm's unit sphere](#) · [Absolute maximization forces a near-maximal base](#) · [Continuous linear maps bounded by a seminorm](#) · [Maximal minors scaled by the reference ball volume](#) · [The recovered contraction maps one unit ball onto the other](#) · [Each satellite norms its coefficient preimage in absolute value](#) · [A nearby norming point gives an almost-norming evaluation](#)

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Level 0

Inverse maps on open sets with global metric bounds

Records the maps, their inverse relations, and the quantitative bounds used in the orientation argument.

`MathlibAnnex.BilipschitzOrientation.BilipschitzOpenData`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A positive lower metric bound passes to the derivative](#) · [Weak Piola identity for a compactly supported test field](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

An attained absolute determinant maximum

Selects a maximizing dual frame for a specified basis.

`MathlibAnnex.DeterminantFrame.exists_maximizingFrame`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Absolute maximization forces a near-maximal base](#) · [The set of near-maximal dual frames](#) · [Positivity of the determinant maximum](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Equality of seminorm balls under a linear bijection preserves the seminorms

Turns a ball-image equality into one seminorm inequality in each direction.

`MathlibAnnex.SeminormBall.map_eq`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [All Plücker bodies determine the norm up to linear isometry](#)

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Level 0

Identifying constants on an open overlap

Uses positive measure to find a point where both almost-everywhere equalities hold.

`MathlibAnnex.aeConstants_eq_of_open_overlap`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [From local to global almost-everywhere constancy](#)

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Level 0

Differentiating a local mollification through its kernel

Writes a directional derivative without differentiating the locally integrable input.

`MathlibAnnex.WeakGradient.localMollification_fderiv_apply`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Zero weak gradient gives zero derivatives of local mollifications](#)

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Level 1 (18 Cards)

Level 1

Absolute maximization forces a near-maximal base

A dominant determinant term prevents a large deficit in the base frame.

MathlibAnnex.Satellite.absoluteMaximizer_base_nearMax

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#) · [A weighted determinant polynomial with satellites](#) · [An attained absolute determinant maximum](#)

Used by in this scope: [The chosen support slice consists of almost-isometric generators](#) · [A finite absolute-maximizing family almost norms the unit sphere](#)

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Level 1

Positivity of the determinant maximum

A uniformly scaled dual basis supplies a positive comparison determinant.

MathlibAnnex.DeterminantFrame.determinantMaximum_pos

Immediate Card prerequisites: [An attained absolute determinant maximum](#)

Used by in this scope: [A common inverse estimate from row Cramer](#)

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Level 1

Strong local convergence of mollified derivatives

Approximates the derivative in local L^n , where n is the dimension of the domain.

MathlibAnnex.Mollification.tendsto_eLpNorm_fderiv_mollify_sub

Immediate Card prerequisites: [Mollification by a normalized smooth kernel](#)

Used by in this scope: [Maximal-minor integral differences for Lipschitz perturbations](#)

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Level 1

The divergence-free cofactor identity

Cancels the Hessian terms that arise when differentiating a cofactor row.

MathlibAnnex.Piola.divergence_cofactorRowField_eq_zero

Immediate Card prerequisites: [A cofactor row by row replacement](#)

Used by in this scope: [A component flux gives a determinant difference](#)

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Level 1

Continuous linear maps bounded by a seminorm

Defines the pointwise contraction condition relative to a stored equivalent seminorm.

`MathlibAnnex.EquivalentSeminorm.IsContraction`

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#)

Used by in this scope: [Compactness of the signed generator set](#) · [The symmetric convex body of contraction minors](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

A strict lower bound on a norm's unit sphere

Defines the strict lower condition on a fixed norm unit sphere.

`MathlibAnnex.FiniteSup.FinMapAlmostIsometric`

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#)

Used by in this scope: [The chosen support slice consists of almost-isometric generators](#)

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Level 1

Vanishing weak gradient tested by divergence

Expresses a weak equation through compactly supported continuously differentiable vector fields.

`MathlibAnnex.WeakDivergenceZero`

Immediate Card prerequisites: [Divergence as the trace of a derivative](#)

Used by in this scope: [Zero weak gradient gives zero derivatives of local mollifications](#) · [The transported Jacobian sign has zero weak gradient](#)

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Level 1

The satellite polynomial is linear in the maximal-minor vector

Expresses the configuration polynomial as a linear pairing with maximal minors.

`MathlibAnnex.PluckerSupport.pluckerPairing_satelliteCoefficients_maximalMinors`

Immediate Card prerequisites: [The family of maximal minors in increasing row order](#) · [A weighted determinant polynomial with satellites](#)

Used by in this scope: [The chosen support slice consists of almost-isometric generators](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

Maximal minors scaled by the reference ball volume

Fixes the volume factor and increasing-row signs.

`MathlibAnnex.Matrix.ballVolumeScaledMaximalMinors`

Immediate Card prerequisites: [The family of maximal minors in increasing row order](#) · [A seminorm with two-sided bounds against a reference norm](#)

Used by in this scope: [Compactness of the signed generator set](#) · [The symmetric convex body of contraction minors](#) · [The set average of derivative generators](#)

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Level 1

A positive lower metric bound passes to the derivative

Preserves the same lower constant in the difference-quotient limit.

`MathlibAnnex.BilipschitzOrientation.fderiv_lower_bound`

Immediate Card prerequisites: [Inverse maps on open sets with global metric bounds](#)

Used by in this scope: [Signed transfer from the absolute Jacobian formula](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

The set of near-maximal dual frames

Records a determinant deficit while keeping all dual rows contractive.

`MathlibAnnex.DeterminantFrame.nearMaxFrames`

Immediate Card prerequisites: [An attained absolute determinant maximum](#)

Used by in this scope: [A common inverse estimate from row Cramer](#) · [Each satellite norms its coefficient preimage in absolute value](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

From local to global almost-everywhere constancy

Separates agreement of constants by connectedness from countable measure-theoretic gluing.

`MathlibAnnex.exists_aeConstantOn_of_local`

Immediate Card prerequisites: [Gluing one almost-everywhere constant over a countable cover](#) · [Identifying constants on an open overlap](#)

Used by in this scope: [Global almost-everywhere constancy on a preconnected domain](#)

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Level 1

The radial extension is 3-Lipschitz

Separates radial variation from the change of unit direction to obtain the global bound with coefficient $1 + 2$.

`MathlibAnnex.Sphere.lipschitzWith_radialExtension`

Immediate Card prerequisites: [Radial extension of an isometry between unit spheres](#)

Used by in this scope: [Finite-dimensionality transfers across a sphere isometry](#) · [Isometric unit spheres force equal finite dimensions](#) · [A common orientation sign for the radial derivative average](#)

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Level 1

The inverse sphere isometry gives the inverse radial map

Shows that extending the inverse sphere isometry radially undoes the forward radial extension.

`MathlibAnnex.Sphere.radialExtension_leftInverse`

Immediate Card prerequisites: [Radial extension of an isometry between unit spheres](#)

Used by in this scope: [Finite-dimensionality transfers across a sphere isometry](#) · [Isometric unit spheres force equal finite dimensions](#) · [A common orientation sign for the radial derivative average](#)

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Level 1

A nearby norming point gives an almost-norming evaluation

Two triangle inequalities transfer absolute norm attainment to a unit vector.

`MathlibAnnex.Satellite.abs_eval_lower_of_norms_nearby`

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#)

Used by in this scope: [One fixed satellite family almost norms every detected unit vector](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

The recovered contraction maps one unit ball onto the other

Upgrades ball inclusion to equality by strict compact-subset volume comparison.

`MathlibAnnex.PluckerRecovery.LimitCertificate.image_unitBall_eq`

Immediate Card prerequisites: [A compact proper subset of a seminorm ball has smaller Haar measure](#) · [A seminorm with two-sided bounds against a reference norm](#)

Used by in this scope: [All Plücker bodies determine the norm up to linear isometry](#)

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Level 1

Determinant integrals under strong L^n convergence

Uses Hölder's inequality to turn strong L^n convergence of matrix fields into convergence of their determinant integrals.

`MathlibAnnex.MeasureTheory.tendsto_integral_det_of_strongLn`

Immediate Card prerequisites: [A determinant difference bound in the sup operator norm](#)

Used by in this scope: [Maximal-minor integral differences for Lipschitz perturbations](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

A common generator selected from equal convex hulls

Uses a first maximizing slice and a separating sequence to obtain an original common point carrying a prescribed property.

`MathlibAnnex.exists_common_of_convexHull_eq`

Immediate Card prerequisites: [Successive maximum refinement by an ordered list of functionals](#) · [A support face is the convex hull of the maximizing generators](#)

Used by in this scope: [Equal Plücker bodies yield matching generators with an almost-isometric representative](#) · [Almost-contractive linear recovery with exact volume scaling](#)

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Level 2 (10 Cards)

Level 2

Compactness of the signed generator set

Passes compactness from contractions to their two signed minor images.

`MathlibAnnex.PluckerBody.isCompact_generators`

Immediate Card prerequisites: [Continuous linear maps bounded by a seminorm](#) · [Maximal minors scaled by the reference ball volume](#)

Used by in this scope: [The chosen support slice consists of almost-isometric generators](#) · [The average of contractive derivative generators lies in the body](#)

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Level 2

The set average of derivative generators

Separates the volume-scaled integrand from the normalization of its average.

`MathlibAnnex.Plucker.derivativeAverage`

Immediate Card prerequisites: [Maximal minors scaled by the reference ball volume](#)

Used by in this scope: [The average of contractive derivative generators lies in the body](#) · [A common orientation sign for the radial derivative average](#)

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Level 2

The symmetric convex body of contraction minors

Takes the real convex hull of both signs of every contraction generator.

`MathlibAnnex.PluckerBody.body`

Immediate Card prerequisites: [Continuous linear maps bounded by a seminorm](#) · [Maximal minors scaled by the reference ball volume](#)

Used by in this scope: [Equal Plücker bodies yield matching generators with an almost-isometric representative](#) · [The average of contractive derivative generators lies in the body](#) · [Almost-contractive linear recovery with exact volume scaling](#)

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Level 2

Isometric unit spheres force equal finite dimensions

Compares Hausdorff dimensions in both directions using the surjective Lipschitz radial maps.

`MathlibAnnex.Sphere.finrank_eq`

Immediate Card prerequisites: [The inverse sphere isometry gives the inverse radial map](#) · [The radial extension is 3-Lipschitz](#)

Used by in this scope: [The unit-sphere chord metric determines the ambient normed space](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

A common inverse estimate from row Cramer

Controls all near-maximal inverse frames by an explicit basis-dependent constant.

`MathlibAnnex.DeterminantFrame.inverseBoundConstant_bound`

Immediate Card prerequisites: [The set of near-maximal dual frames](#) · [Positivity of the determinant maximum](#)

Used by in this scope: [A finite absolute-maximizing family almost norms the unit sphere](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

A component flux gives a determinant difference

Expresses the change from one output-coordinate perturbation as a divergence.

`MathlibAnnex.Piola.divergence_componentFlux_eq_det_sub`

Immediate Card prerequisites: [The divergence-free cofactor identity](#)

Used by in this scope: [Smooth compact perturbations preserve the determinant integral difference](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

Zero weak gradient gives zero derivatives of local mollifications

Substitutes a translated smooth kernel into the weak equation and tracks the reflection sign.

`MathlibAnnex.WeakGradient.localMollification_fderiv_apply_eq_zero`

Immediate Card prerequisites: [Vanishing weak gradient tested by divergence](#) · [Differentiating a local mollification through its kernel](#)

Used by in this scope: [Local almost-everywhere constancy from the weak equation](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

Signed transfer from the absolute Jacobian formula

Transfers an integrable function without first assuming that the Jacobian sign is constant.

`MathlibAnnex.BilipschitzOrientation.signed_area_transfer`

Immediate Card prerequisites: [A positive lower metric bound passes to the derivative](#)

Used by in this scope: [The transported Jacobian sign has zero weak gradient](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

Finite-dimensionality transfers across a sphere isometry

Uses the radial homeomorphism to transfer local compactness, then applies the finite-dimensionality criterion for real normed spaces.

`MathlibAnnex.Sphere.finiteDimensional_codomain`

Immediate Card prerequisites: [The inverse sphere isometry gives the inverse radial map](#) · [The radial extension is 3-Lipschitz](#)

Used by in this scope: [The unit-sphere chord metric determines the ambient normed space](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

Each satellite norms its coefficient preimage in absolute value

Varying one functional at an absolute maximizer forces endpoint attainment.

`MathlibAnnex.Satellite.absoluteMaximizer_satellite_norms_preimage`

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#) · [A weighted determinant polynomial with satellites](#) · [The set of near-maximal dual frames](#)

Used by in this scope: [One fixed satellite family almost norms every detected unit vector](#)

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Level 3 (4 Cards)

Level 3

The average of contractive derivative generators lies in the body

Uses closed-convex average membership for the actual derivative-generator function.

`MathlibAnnex.Plucker.derivativeAverage_mem_body`

Immediate Card prerequisites: [Compactness of the signed generator set](#) · [The symmetric convex body of contraction minors](#) · [The set average of derivative generators](#) · [The convex hull of a compact set in a finite real coordinate space](#)

Used by in this scope: [One boundary extension with all five analytic properties](#)

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Level 3

Smooth compact perturbations preserve the determinant integral difference

Builds a finite telescope from single-output perturbations with compactly supported fluxes.

`MathlibAnnex.Piola.integral_det_fderiv_add_sub_eq_zero_of_contDiff`

Immediate Card prerequisites: [A component flux gives a determinant difference](#) · [Zero integral of a compactly supported divergence](#)

Used by in this scope: [Maximal-minor integral differences for Lipschitz perturbations](#)

[Open Declaration Card](#) [Read exact source](#)

Level 3

Local almost-everywhere constancy from the weak equation

Turns constant mollifications into one almost-everywhere constant by choosing a common convergence point.

`MathlibAnnex.WeakDivergenceZero.nonempty_localAECOnstantAt`

Immediate Card prerequisites: [Zero weak gradient gives zero derivatives of local mollifications](#)

Used by in this scope: [Global almost-everywhere constancy on a preconnected domain](#)

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Level 3

One fixed satellite family almost norms every detected unit vector

Uses a detecting coefficient and its matching satellite without replacing the family.

`MathlibAnnex.Satellite.absoluteMaximizer_satellites_almost_norm`

Immediate Card prerequisites: [Each satellite norms its coefficient preimage in absolute value](#) · [A nearby norming point gives an almost-norming evaluation](#)

Used by in this scope: [The chosen support slice consists of almost-isometric generators](#) · [A finite absolute-maximizing family almost norms the unit sphere](#)

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Level 4 (5 Cards)

Level 4

One boundary extension with all five analytic properties

Keeps boundary agreement, contraction, derivative control, integrability and average membership on one witness.

`MathlibAnnex.Plucker.exists_extension_with_derivativeAverage_mem`

Immediate Card prerequisites: [A seminorm increment bound passes to every derivative direction](#) · [The average of contractive derivative generators lies in the body](#)

Used by in this scope: [A target contraction generator lies in the source body](#)

[Open Declaration Card](#) [Read exact source](#)

Level 4

The chosen support slice consists of almost-isometric generators

Obtains one almost-isometric map from each point of the specified raw maximum slice.

`MathlibAnnex.FiniteRecovery.targetRawSupportMaximizer_good`

Immediate Card prerequisites: [The satellite polynomial is linear in the maximal-minor vector](#) · [Compactness of the signed generator set](#) · [A strict lower bound on a norm's unit sphere](#) · [Absolute maximization forces a near-maximal base](#) · [One fixed satellite family almost norms every detected unit vector](#)

Used by in this scope: [Equal Plücker bodies yield matching generators with an almost-isometric representative](#) · [Almost-contractive linear recovery with exact volume scaling](#)

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Level 4

A finite absolute-maximizing family almost norms the unit sphere

Chooses the inverse bound, net and weight before fixing one satellite family.

`MathlibAnnex.Satellite.exists_goodSatellitePackage`

Immediate Card prerequisites: [A common inverse estimate from row Cramer](#) · [Absolute maximization forces a near-maximal base](#) · [One fixed satellite family almost norms every detected unit vector](#)

Used by in this scope: [Equal Plücker bodies yield matching generators with an almost-isometric representative](#) · [Almost-contractive linear recovery with exact volume scaling](#)

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Level 4

Global almost-everywhere constancy on a preconnected domain

Combines local weak-gradient rigidity with topological and countable gluing.

`MathlibAnnex.WeakDivergenceZero.exists_aeConstantOn`

Immediate Card prerequisites: [Local almost-everywhere constancy from the weak equation](#) · [From local to global almost-everywhere constancy](#)

Used by in this scope: [Almost-everywhere constancy of the sign on a preconnected target](#)

[Open Declaration Card](#) [Read exact source](#)

Level 4

Maximal-minor integral differences for Lipschitz perturbations

Passes the smooth compact-perturbation identity to Lipschitz maps through local strong convergence.

`MathlibAnnex.NullLagrangian.integral_maximalMinor_fderiv_add_sub_eq_zero_of_lipschitzWith`

Immediate Card prerequisites: [Determinant integrals under strong \$L^n\$ convergence](#) · [Strong local convergence of mollified derivatives](#) · [Smooth compact perturbations preserve the determinant integral difference](#)

Used by in this scope: [Boundary agreement determines maximal-minor integrals](#) · [Weak Piola identity for a compactly supported test field](#)

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Level 5 (4 Cards)

Level 5

Almost-contractive linear recovery with exact volume scaling

Recovers one bijective linear factor with a norm estimate and exact volume scaling.

`MathlibAnnex.PluckerRecovery.nonempty_linearCertificate_of_pluckerBodies_eq`

Immediate Card prerequisites: [The chosen support slice consists of almost-isometric generators](#) · [The symmetric convex body of contraction minors](#) · [Cramer's rule for coordinates of a row](#) · [A finite absolute-maximizing family almost norms the unit sphere](#) · [A common generator selected from equal convex hulls](#)

Used by in this scope: [A volume-normalized linear contraction obtained by a limit](#)

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Level 5

Equal Plücker bodies yield matching generators with an almost-isometric representative

Selects common minor coordinates with a target almost-isometric representative.

`MathlibAnnex.FiniteRecovery.nonempty_almostIsometryMatch_of_all_body_eq`

Immediate Card prerequisites: [The chosen support slice consists of almost-isometric generators](#) · [The symmetric convex body of contraction minors](#) · [A finite absolute-maximizing family almost norms the unit sphere](#) · [A common generator selected from equal convex hulls](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Level 5

Boundary agreement determines maximal-minor integrals

Turns pointwise equality on a seminorm sphere into a compactly supported Lipschitz perturbation.

`MathlibAnnex.NullLagrangian.integral_maximalMinor_eq_of_pointwise_boundary_eq`

Immediate Card prerequisites: [Maximal-minor integral differences for Lipschitz perturbations](#)

Used by in this scope: [A target contraction generator lies in the source body](#)

[Open Declaration Card](#) [Read exact source](#)

Level 5

Weak Piola identity for a compactly supported test field

Uses coordinate perturbations to show that the signed pullback of a test-field divergence has zero integral.

`MathlibAnnex.BilipschitzOrientation.weak_piola`

Immediate Card prerequisites: [Divergence as the trace of a derivative](#) · [Maximal-minor integral differences for Lipschitz perturbations](#) · [Inverse maps on open sets with global metric bounds](#)

Used by in this scope: [The transported Jacobian sign has zero weak gradient](#)

[Open Declaration Card](#) [Read exact source](#)

Level 6 (2 Cards)

Level 6

A volume-normalized linear contraction obtained by a limit

Takes a compact operator limit preserving the exact volume equation.

`MathlibAnnex.PluckerRecovery.nonempty_limitCertificate_of_pluckerBodies_eq`

Immediate Card prerequisites: [Almost-contractive linear recovery with exact volume scaling](#)

Used by in this scope: [All Plücker bodies determine the norm up to linear isometry](#)

[Open Declaration Card](#) [Read exact source](#)

Level 6

The transported Jacobian sign has zero weak gradient

Uses signed transfer and weak Piola with the same test field to obtain the weak equation on the target.

`MathlibAnnex.BilipschitzOrientation.weakDivergenceZero_targetJacobianSign`

Immediate Card prerequisites: [Vanishing weak gradient tested by divergence](#) · [Signed transfer from the absolute Jacobian formula](#) · [Weak Piola identity for a compactly supported test field](#)

Used by in this scope: [Almost-everywhere constancy of the sign on a preconnected target](#)

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Level 7 (2 Cards)

Level 7

All Plücker bodies determine the norm up to linear isometry

Recovers a real linear equivalence preserving the two supplied norms.

`MathlibAnnex.EquivalentSeminorm.nonempty_linearIsometryEquiv_of_pluckerBodies_eq`

Immediate Card prerequisites: [Equality of seminorm balls under a linear bijection preserves the seminorms](#) · [A volume-normalized linear contraction obtained by a limit](#) · [The recovered contraction maps one unit ball onto the other](#)

Used by in this scope: [The unit-sphere chord metric determines the ambient normed space](#)

[Open Declaration Card](#) [Read exact source](#)

Level 7

Almost-everywhere constancy of the sign on a preconnected target

Obtains one constant from the weak equation and preconnectedness of the target.

`MathlibAnnex.BilipschitzOrientation.targetJacobianSign_ae_const`

Immediate Card prerequisites: [Global almost-everywhere constancy on a preconnected domain](#) · [The transported Jacobian sign has zero weak gradient](#)

Used by in this scope: [The signed Jacobian integral is a sign times target volume](#)

[Open Declaration Card](#) [Read exact source](#)

Level 8 (1 Card)

Level 8

The signed Jacobian integral is a sign times target volume

Uses preconnectedness to obtain a constant, positive measure to identify its sign, and finite measure to integrate it.

`MathlibAnnex.BilipschitzOrientation.integral_det_fderiv_eq_signed_volume`

Immediate Card prerequisites: [Almost-everywhere constancy of the sign on a preconnected target](#)

Used by in this scope: [A common orientation sign for the radial derivative average](#)

[Open Declaration Card](#) [Read exact source](#)

Level 9 (1 Card)

Level 9

A common orientation sign for the radial derivative average

Identifies the whole minor vector using one signed Jacobian integral.

`MathlibAnnex.Plucker.derivativeAverage_comp_linear_radial`

Immediate Card prerequisites: [The inverse sphere isometry gives the inverse radial map](#) · [The set average of derivative generators](#) · [The signed Jacobian integral is a sign times target volume](#) · [The radial extension is 3-Lipschitz](#)

Used by in this scope: [A target contraction generator lies in the source body](#)

[Open Declaration Card](#) [Read exact source](#)

Level 10 (1 Card)

Level 10

A target contraction generator lies in the source body

Transfers membership through two extensions with the same boundary trace.

`MathlibAnnex.PluckerBody.normalizedGenerator_mem_of_sphereIsometry`

Immediate Card prerequisites: [One boundary extension with all five analytic properties](#) · [Boundary agreement determines maximal-minor integrals](#) · [A common orientation sign for the radial derivative average](#)

Used by in this scope: [Sphere isometries preserve the Plücker body](#)

[Open Declaration Card](#) [Read exact source](#)

Level 11 (1 Card)

Level 11

Sphere isometries preserve the Plücker body

Combines generator transport and its inverse for every finite output dimension.

`MathlibAnnex.PluckerBody.eq_of_sphereIsometry`

Immediate Card prerequisites: [A target contraction generator lies in the source body](#)

Used by in this scope: [The unit-sphere chord metric determines the ambient normed space](#)

[Open Declaration Card](#) [Read exact source](#)

Level 12 (1 Card)

Level 12

The unit-sphere chord metric determines the ambient normed space

Classifies existence of ambient linear isometries from the unit-sphere chord metric.

`MathlibAnnex.Sphere.nonempty_isometryEquiv_iff_nonempty_linearIsometryEquiv_of_finiteDimensional`

Immediate Card prerequisites: [Finite-dimensionality transfers across a sphere isometry](#) · [Isometric unit spheres force equal finite dimensions](#) · [All Plücker bodies determine the norm up to linear isometry](#) · [Sphere isometries preserve the Plücker body](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)